Om Prakash Singh

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Professional Summary

SoC Validation Engineer with 8 years in pre- and post-silicon validation — CPU power management (PM), bring-up, and cross-IP debug. Hands-on with bare-metal C/C++ test development, FPGA-based emulation (Zebu), JTAG, and protocol analyzers. Uses agentic AI workflows to accelerate trace analysis and debug. Experienced in defining validation strategy, translating architecture and PM specs into coverage plans at subsystem/SoC levels, and debugging complex issues across firmware and power domains.

Experience

Device Validation Engineer 2 (System-Level Validation) *Hyderabad, India*
Microsoft *June 2024 – Present*

- Validate AI models across IHV silicon (Intel, AMD, Qualcomm) on Copilot+ PCs — capture and analyze KPIs for performance, accuracy, and memory, verifying correctness across hardware vendors
- Built EdgeOptimizer to capture per-workload KPIs (performance, accuracy, memory) during AI model execution on real silicon, enabling cross-IHV comparison and regression tracking
- Drove RCA on WPR/WPA hardware traces and ETL logs; partnered with silicon vendors to triage scheduling, concurrency, and suspected silicon issues — deployed agentic AI workflows to parse and correlate trace data, reducing debug cycle time
- Validated multi-model concurrency with semaphore-based execution control; debugged scheduling and resource-contention issues across CPU/NPU to verify deterministic KPI capture under load

Emulation Engineer (Pre-Silicon Validation) *Bangalore, India*
Intel *July 2021 – June 2024*

- Owned pre-silicon CPU power-management validation — C-state entry/exit, ADR, and P-channel handshakes — on Xeon server SoCs; translated CPU architecture and PM specs into coverage plans and drove silicon-readiness milestones
- Developed 10+ PCIe Gen6 transactor enhancements in C++ and validation content using Perspec on Zebu (FPGA) emulation; debugged 15+ cross-IP issues spanning clock/power gating, reset sequencing, and PM handshakes using bare-metal C
- Represented validation in cross-functional CPU architecture and power-management reviews; influenced PM flow definitions and contributed to ˜25% reduction in post-silicon escapes
- Built reusable Perspec validation content and regression checks for CPU PM and PCIe scenarios on Zebu, improving coverage repeatability across SoC readiness milestones

Software Engineer (Post-Silicon Validation) *Bangalore, India*
L&T Technology Services *Sep 2018 – July 2021*

- Contributed to post-silicon bring-up and validation of NVMe storage controller SoCs (SanDisk) from power-on through production, using JTAG (Metaware) and Lecroy protocol analyzers
- Developed Python automation covering 25+ power management test scenarios; helped debug power-related silicon issues through cross-functional firmware collaboration
- Validated PCIe/NVMe protocol compliance and power state transitions; supported SDExpress POC validation as part of the team

Education

- **M.Tech in Computer Science & Engineering**, National Institute of Technology, Durgapur – 2018
- **B.Tech in Computer Science & Engineering**, RGPV, Bhopal – 2015

Technical Skills

- **Programming & Scripting:** C, C++, Python, Linux
- **Validation & Debug:** Pre-/Post-Silicon Validation, CPU, SoC Bring-Up, JTAG, Protocol Debug
- **Systems & Architecture:** CPU Architecture, CPU Power Management (C-states), SoC, PCIe/NVMe
- **Tools:** Verdi, JTAG (Metaware), Lecroy Analyzer, Oscilloscope, WPR/WPA
- **AI & Agentic Workflows:** Agentic AI, GitHub Copilot CLI, ONNX Runtime
- **Domains:** Silicon Validation, Storage/NVMe, PCIe

Awards & Recognition

- Intel: Star of the Month for CXL firmware validation and early bug discovery
- L&T/SanDisk: Spot Awards for SDExpress POC validation